Proofs From the Ground Up

A Deep Conversational Reader Before Velleman

Engineering Mathematics to Pure Mathematics

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How This Book Should Be Read

This is not a survey. A survey flies over the land and names the rivers. This reader is meant to put your boots in the mud. We will spend a long time on the first ideas because the first ideas carry everything later. If statements, quantifiers, definitions, sets, and functions are fuzzy, then real analysis feels like weather, abstract algebra feels like arbitrary rules, and topology feels like someone replaced geometry with riddles.

So we will not rush.

The path is the same one you asked for: from basic logic and notation toward the proof-writing maturity needed for Velleman, then onward toward analysis, algebra, topology, differential geometry, and algebraic geometry. But the emphasis here is on depth before motion. Each chapter tries to give the same concept from several angles:

- the informal picture,
- the formal definition,
- examples and non-examples,
- what the notation is really saying,
- the common beginner mistakes,
- historical and later-field connections,
- worked proofs,
- exercises you can write,
- questions you can think through while walking.

Mathematics is not made of symbols. It is made of ideas disciplined by symbols. The discipline matters. So does the idea.

Focus

Do not measure progress by pages finished. Measure it by whether you can close the PDF and explain the idea without pretending.

The rule for this reader

When a definition appears, do not merely nod. Work it. Produce examples. Break it. Repair it. Ask why every word is present. Good mathematicians do this almost automatically. Beginners often think they are supposed to skip it and get to the theorem. That is backwards. The theorem is built out of definitions.

What “beginner friendly” means here

Beginner friendly does not mean vague. It means no step is treated as beneath explanation. We will be conversational, but we will end in exact statements. The point is not to make rigorous mathematics soft. The point is to let rigor arrive with enough handles that you can hold it.

1 The Change in Game: From Getting Answers to Giving Reasons

Engineering mathematics often trains you to move quickly from problem type to method. That training is valuable. If you see a linear system, you think matrices. If you see a differential equation, you scan for separation, integrating factors, characteristic equations, transforms. If you see an integral, you try substitution, integration by parts, symmetry, or numerical approximation.

Pure mathematics asks you to slow down at a different point. Before asking "what method solves this?", it asks:

> What exactly is being claimed, and why does it follow from the assumptions?

That question sounds modest. It is not. It changes what counts as success.

1.1 A calculation can be right for the wrong reason

Suppose you notice:

$$1 = 1^2, \quad 1+3 = 4 = 2^2, \quad 1+3+5 = 9 = 3^2, \quad 1+3+5+7 = 16 = 4^2.$$

It is natural to guess:

$$1 + 3 + 5 + \cdots + (2n-1) = n^2.$$

You can check this for $n = 5, 6, 7, 100$. You can even write a program to check it for the first million values. That is useful evidence. It is not proof.

Why not? Because the claim is not about the first million natural numbers. It is about every natural number. A proof is the device that crosses that gap.

Intuition

A computation samples the landscape. A proof explains the geology.

1.2 What proof adds

A proof gives a reason the statement cannot fail. For the odd-number identity, one proof is visual: build a square by adding L-shaped borders of size $1, 3, 5, \ldots$. Another proof is induction. Another uses the arithmetic series formula. The same theorem can have several proofs because the theorem has several inner explanations.

This matters for mathematical maturity. Mature readers do not just ask whether a proof is correct. They ask what the proof reveals.

1.3 From formulas to structures

Engineering math often focuses on formulas:

$$F = ma, \qquad e^{i\theta} = \cos\theta + i\sin\theta, \qquad A\vec{x} = \vec{b}.$$

Pure math often asks about structures:

- What is a function?
- What is a set?
- What is an operation?
- What is a space?
- What does it mean for two structures to be the same?
- What is preserved by a map?

This is not abstraction for sport. It is compression. One theorem about groups can explain clock arithmetic, symmetries of a square, permutations, and invertible matrices. One theorem about compactness can explain why continuous functions behave well on closed intervals and why certain optimization problems have solutions.

History and connection

Many abstractions were invented after concrete methods succeeded too well. Calculus worked before its foundations were clean. Algebra solved equations before it became the study of structure. Geometry measured shapes before topology asked what "same shape" could mean without rulers. The rigor came because the ideas were powerful enough to need a stronger language.

1.4 Exercise: three kinds of understanding

For the identity

$$1 + 3 + \cdots + (2n-1) = n^2,$$

write three explanations:

1. a computational check for $n = 5$,
2. an informal visual explanation,

3. a proof plan for induction.

Do not worry yet about perfect formal style. Notice how each explanation gives a different kind of confidence.

Walking question

Think of a formula you know well from engineering. Ask: do I know why it is true, where it fails, and which assumptions make it work?

2 Mathematical Sentences: Statements, Predicates, and Claims

Before proof, there is grammar. Not school grammar, but mathematical grammar: what kind of sentence is this, what objects does it mention, and can it be true or false?

Definition 1. A *statement* is a sentence that is either true or false.

Examples:

- "7 is prime" is a true statement.
- "9 is prime" is a false statement.
- "There are infinitely many primes" is a true statement.
- "Every continuous function is differentiable" is a false statement.

Notice that a statement does not have to be easy. For centuries, Fermat's Last Theorem was a statement whose truth was unknown. It was still a statement.

2.1 Not every mathematical-looking phrase is a statement

The expression

$$x^2 + 1$$

is not a statement. It does not assert anything.

The sentence

$$x^2 + 1 = 10$$

is not a statement until x has a value or a quantifier. It is a predicate.

The command

> Solve $x^2 + 1 = 10$.

is not a statement either. It is an instruction.

The question

> Is $x^2 + 1 = 10$?

is not a statement. It asks for one.

Definition 2. A *predicate* is a sentence with one or more variables that becomes a statement when the variables are assigned values from a specified domain.

2.2 Predicates are like Boolean-valued functions, with a warning

In software, you might write:

```
function isPrime(n) {
  ...
}
```

The expression `isPrime(n)` is not a value until `n` is supplied. In mathematics, $P(n)$ behaves similarly. But mathematics forces you to say what kind of input n is allowed to be. Natural number? Integer? Real? Polynomial? Matrix?

The predicate

$$P(x) : x^2 < 2$$

has different flavor depending on the domain:

- over $\mathbb{N}$, it is true only for $x = 1$ if $\mathbb{N} = \{1, 2, \ldots\}$,
- over $\mathbb{Z}$, it is true for $x = -1, 0, 1$,
- over $\mathbb{R}$, it is true for $-\sqrt{2} < x < \sqrt{2}$.

The same symbols do not always describe the same mathematical situation.

2.3 The beginner mistake: reading predicates as statements

A common early error is to see something like

$$x^2 \geq 0$$

and treat it as automatically true. But true where? If $x \in \mathbb{R}$, yes. If $x \in \mathbb{C}$, the symbol $\geq$ is not even naturally defined for complex numbers in the same way. The domain matters.

Common trap

Never let a variable float without a domain. A variable without a domain is a character without a story.

2.4 Claims

A theorem, proposition, lemma, or corollary is a statement that is intended to be proved. The names mostly indicate role and importance.

- A *theorem* is a major result.

- A *proposition* is a result of moderate importance.
- A *lemma* is a helper result used to prove something else.
- A *corollary* follows quickly from a previous result.

These are cultural labels, not rigid species. One book's lemma may be another book's theorem.

2.5 Exercises

Classify each as statement, predicate, expression, command, or question.

1. $3 + 4 = 7$.
2. $x + 4 = 7$.
3. Find all x such that $x + 4 = 7$.
4. $\sin^2 x + \cos^2 x$.
5. Every finite group of prime order is cyclic.
6. Is every ring a field?
7. $A \subseteq B$.
8. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous.

For every predicate, state at least two possible domains and explain how the meaning changes.

3 Variables Have Roles: Fixed, Arbitrary, Chosen, Bound

Beginners often treat variables as generic letters. In proofs, variables have jobs. Confusing those jobs is one of the fastest ways to get lost.

3.1 Fixed but unknown

When a proof says:

> Let $n \in \mathbb{Z}$.

we introduce one integer n. We may not know whether it is 5 or -20 or 0, but for the duration of the argument it is fixed.

This is like receiving an input to a function. You do not know the value, but the value is fixed during that run.

3.2 Arbitrary

When a proof says:

> Let $n \in \mathbb{Z}$ be arbitrary.

the word arbitrary means we are not allowed to use any special property of n beyond $n \in \mathbb{Z}$. If the proof works for this arbitrary n, it works for all integers.

This is how universal statements are proved.

3.3 Chosen as a witness

When proving

$$\exists n \in \mathbb{Z} \text{ such that } n^2 = 49,$$

you may choose $n = 7$. That chosen object is called a witness.

Existential proofs often turn on finding the right witness. Sometimes the witness is explicit, like 7. Sometimes it is constructed from earlier information. Sometimes existence is proved indirectly, which can be unsettling at first.

3.4 Bound variables

In the statement

$$\forall x \in \mathbb{R},\ x^2 \geq 0,$$

the variable x is bound by the quantifier $\forall$. It lives inside that sentence.

In

$$\sum_{i=1}^{n} i^2,$$

the variable i is bound by the summation. It is a local variable.

This is scope. If you have programmed, you already know the smell of a scope error. Mathematics has them too.

3.5 Free variables

In

$$x^2 \geq 0,$$

the variable x is free unless earlier context has defined it.

Free variables are not bad, but they require context. A theorem statement should not leave important variables floating ambiguously.

3.6 Worked parsing

Consider:

$$\forall x \in \mathbb{R},\ \exists y \in \mathbb{R} \text{ such that } y > x.$$

The variable x is chosen arbitrarily from $\mathbb{R}$. After x is known, the statement promises that some y can be found. A proof can say:

> Let $x \in \mathbb{R}$ be arbitrary. Choose $y = x + 1$. Then $y \in \mathbb{R}$ and $y > x$.

Here x is arbitrary, while y is a witness depending on x.

Common trap

Do not choose a witness before the quantifiers allow it. In $\forall x\, \exists y$, y may depend on x. In $\exists y\, \forall x$, it may not.

3.7 Exercises

For each statement, identify which variables are arbitrary and which are witnesses.

1. $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}$ such that $n = 2m$.
2. $\exists m \in \mathbb{Z}$ such that $\forall n \in \mathbb{Z},\ n = 2m$.
3. $\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N,\ |a_n - L| < \varepsilon$.
4. $\forall a \in A, \forall b \in B, \exists c \in C$ such that $F(a, b, c)$.

Walking question

When you hear a theorem, ask: which objects does the world hand me, and which objects am I allowed to choose in response?

4 Quantifiers: The Game of For Every and There Exists

The symbols $\forall$ and $\exists$ are small, but they decide the shape of the proof.

$$\forall = \text{for every}, \qquad \exists = \text{there exists}.$$

4.1 Universal statements

A universal statement has the form:

$$\forall x \in D,\ P(x).$$

It says every object in the domain D satisfies property P.

To prove it:

> Let $x \in D$ be arbitrary. Show $P(x)$.

To disprove it:

> Find one $x \in D$ such that $P(x)$ is false.

4.2 Existential statements

An existential statement has the form:

$$\exists x \in D \text{ such that } P(x).$$

It says at least one object in D satisfies P.

To prove it:

> Produce a witness $x \in D$ and verify $P(x)$.

To disprove it:

> Show every $x \in D$ fails $P(x)$.

4.3 The quantifier game

Think of $\forall$ as the challenger choosing an object, and $\exists$ as you responding with an object.

Statement:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \text{ such that } x + y = 0.$$

The challenger chooses x. You respond with $y = -x$. You win.

Statement:

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R} \text{ such that } x + y = 0.$$

You must choose y first. Then the challenger chooses x. No single y can cancel every x. You lose.

4.4 A physical analogy

Imagine a thermostat specification:

> For every room temperature target, there exists a control setting that reaches it.

That is plausible. The setting can depend on the target.

Now compare:

> There exists one control setting that reaches every room temperature target.

That is absurd. One setting cannot make the room both 18°C and 24°C at the same time.

4.5 Nested quantifier practice

Statement:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ y > x.$$

True. Given x, choose $y = x + 1$.

Statement:

$$\exists y \in \mathbb{R}, \forall x \in \mathbb{R},\ y > x.$$

False. It asks for a real number larger than every real number.

Statement:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ y^2 = x.$$

False over $\mathbb{R}$, because negative x has no real square root.

True if the domain of x is $[0, \infty)$.

Focus

The domain and the quantifier order often matter more than the formula.

4.6 Exercises

Decide true or false. Prove or give a counterexample.

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ x < y.$
2. $\exists y \in \mathbb{R}, \forall x \in \mathbb{R},\ x < y.$
3. $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z},\ n < m.$
4. $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z},\ n < m.$
5. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R},\ x + y = 10.$
6. $\exists y \in \mathbb{R}, \forall x \in \mathbb{R},\ x + y = 10.$

5 Negation: How to Disagree Precisely

Negation is not a mood. It is a precise operation. To negate a mathematical statement, you must say exactly what it means for the statement to fail.

5.1 Negating universal statements

The negation of

$$\forall x \in D,\ P(x)$$

is

$$\exists x \in D \text{ such that } \neg P(x).$$

If not every object has property P, then at least one object fails property P.

5.2 Negating existential statements

The negation of

$$\exists x \in D \text{ such that } P(x)$$

is

$$\forall x \in D,\ \neg P(x).$$

If no object with property P exists, then every object fails P.

5.3 Everyday versions

Original:

> Every lock in the building opens with this key.

Negation:

> There is at least one lock in the building that does not open with this key.

Original:

> Some server is reachable.

Negation:

> No server is reachable.

5.4 Negating implication

The statement

$$P \Rightarrow Q$$

fails exactly when P is true and Q is false.

So:

$$\neg(P \Rightarrow Q) \iff P \wedge \neg Q.$$

This is important for counterexamples. To disprove:

> If a function is continuous, then it is differentiable.

you need a function that is continuous and not differentiable. The classic example is $f(x) = |x|$ at 0.

5.5 Negating a limit

Definition:

$$a_n \to L$$

means

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N,\ |a_n - L| < \varepsilon.$$

Negation:

$$\exists \varepsilon > 0, \forall N \in \mathbb{N}, \exists n \geq N,\ |a_n - L| \geq \varepsilon.$$

In words: there is some tolerance band around L that the sequence keeps escaping, no matter how far out you go.

History and connection

This kind of quantified precision was part of the nineteenth-century repair of calculus. Mathematicians like Cauchy and Weierstrass wanted to replace hand-wavy infinitesimals and geometric intuition with definitions that could survive pathological examples.

5.6 Exercises

Negate each statement in symbols and plain English.

1. $\forall n \in \mathbb{Z},\ n^2 \geq 0.$
2. $\exists x \in \mathbb{R},\ x^2 + 1 = 0.$

3. $\forall x \in A, \exists y \in B,\ R(x, y)$.
4. $\exists y \in B, \forall x \in A,\ R(x, y)$.
5. If n is prime, then n is odd.
6. Every bounded sequence converges.

Walking question

When you hear a universal claim in everyday life, practice finding the exact counterexample that would refute it.

6 Definitions: The Machines Inside the Words

In ordinary language, definitions can be loose. In mathematics, a definition is an operating manual.

If you know the definition applies, it tells you what you may use. If you want to prove the definition applies, it tells you what you must show.

6.1 Even and odd

Definition 3. An integer n is even if there exists $k \in \mathbb{Z}$ such that $n = 2k$.

Definition 4. An integer n is odd if there exists $k \in \mathbb{Z}$ such that $n = 2k + 1$.

If a proof says n is even, you may write $n = 2k$ for some integer k.

If a proof asks you to show n is even, you must show $n = 2k$ for some integer k.

6.2 Divisibility

Definition 5. For integers a and b, we say a divides b, written $a \mid b$, if there exists $k \in \mathbb{Z}$ such that $b = ak$.

Notice the direction: $a \mid b$ means b is a multiple of a. Beginners often reverse it.

6.3 Opening and closing

Think of a definition as a box.

Opening the box:

> Since $a \mid b$, there exists $k \in \mathbb{Z}$ such that $b = ak$.

Closing the box:

> Since $b = ak$ for some $k \in \mathbb{Z}$, we have $a \mid b$.

Most early proofs are box work.

6.4 Worked proof: even plus even

Theorem 1. *The sum of two even integers is even.*

Proof. Let $a, b \in \mathbb{Z}$ be even. Since a is even, there exists $m \in \mathbb{Z}$ such that $a = 2m$. Since b is even, there exists $n \in \mathbb{Z}$ such that $b = 2n$. Then

$$a + b = 2m + 2n = 2(m + n).$$

Since $m + n \in \mathbb{Z}$, the number $a + b$ has the form $2k$ for an integer k, namely $k = m + n$. Therefore $a + b$ is even. □

6.5 Worked proof: divisibility is transitive

Theorem 2. *If $a, b, c \in \mathbb{Z}$, $a \mid b$, and $b \mid c$, then $a \mid c$.*

Proof. Suppose $a \mid b$ and $b \mid c$. Since $a \mid b$, there exists $m \in \mathbb{Z}$ such that $b = am$. Since $b \mid c$, there exists $n \in \mathbb{Z}$ such that $c = bn$. Substituting,

$$c = bn = (am)n = a(mn).$$

Since $mn \in \mathbb{Z}$, this shows c is an integer multiple of a. Therefore $a \mid c$. □

Common trap

Do not write "$a \mid b$, so $a = bk$." The definition says $b = ak$. The divided object is the one being expressed as a multiple.

6.6 Exercises

Prove each directly from definitions.

1. The product of two even integers is even.
2. The product of two odd integers is odd.
3. If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$.
4. If $a \mid b$, then $a \mid bc$ for every integer c.
5. If a and b are rational, then $a + b$ is rational.

7 If-Then Statements and Proof Strategy

Many theorems are implications:

$$P \Rightarrow Q.$$

The part P is the hypothesis. The part Q is the conclusion.

The proof task is not to show P is true. It is to show that if P is true, then Q must be true.

7.1 Direct proof

Use direct proof when the hypothesis naturally unfolds into the conclusion.

Theorem 3. *If n is divisible by 12, then n is divisible by 3.*

Proof. Suppose n is divisible by 12. Then $n = 12k$ for some $k \in \mathbb{Z}$. Since $12k = 3(4k)$ and $4k \in \mathbb{Z}$, we have $n = 3m$ for the integer $m = 4k$. Therefore n is divisible by 3. □

7.2 Contrapositive

The contrapositive of

$$P \Rightarrow Q$$

is

$$\neg Q \Rightarrow \neg P.$$

These statements are logically equivalent.

Use the contrapositive when the negation of the conclusion is easier to work with than the hypothesis.

Theorem 4. *If n^2 is even, then n is even.*

Proof. We prove the contrapositive: if n is not even, then n^2 is not even. For integers, not even means odd. Suppose n is odd. Then $n = 2k + 1$ for some $k \in \mathbb{Z}$. Thus

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k \in \mathbb{Z}$, n^2 is odd. Hence n^2 is not even. Therefore, by contrapositive, if n^2 is even, then n is even. □

7.3 Contradiction

To prove P by contradiction, assume $\neg P$ and derive an impossibility.

Theorem 5. *There is no smallest positive real number.*

Proof. Suppose, for contradiction, that there is a smallest positive real number r. Since $r > 0$, the number $r/2$ is also positive. But $r/2 < r$, contradicting the claim that r is the smallest positive real number. Therefore no smallest positive real number exists. □

7.4 Choosing the method

Direct proof asks: can I unfold the hypothesis and reach the conclusion?

Contrapositive asks: is the negation of the conclusion more concrete?

Contradiction asks: would denying the result trap me in an impossibility?

Proof lab

For each theorem below, write only the strategy and first sentence.

1. If n^2 is odd, then n is odd.
2. If $a + b$ is irrational and a is rational, then b is irrational.
3. There are infinitely many primes.
4. If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

8 Proof by Contradiction: Power and Danger

Contradiction is dramatic: assume the opposite and show the world breaks. It is also easy to misuse. The goal is not to create confusion. The goal is to force a precise impossibility.

8.1 The irrationality of $\sqrt{2}$

This is one of the classic first proofs because it reveals a hidden structure in the integers: parity and divisibility constrain square roots.

Theorem 6. *$\sqrt{2}$ is irrational.*

Proof. Suppose, for contradiction, that $\sqrt{2}$ is rational. Then $\sqrt{2} = a/b$ for integers a, b with $b \neq 0$. Choose this fraction in lowest terms, so a and b have no common factor greater than 1.

Squaring gives

$$2 = \frac{a^2}{b^2},$$

so

$$a^2 = 2b^2.$$

Thus a^2 is even. By the theorem proved earlier, if a^2 is even, then a is even. So $a = 2k$ for some $k \in \mathbb{Z}$.

Substitute:

$$(2k)^2 = 2b^2.$$

Then

$$4k^2 = 2b^2,$$

so

$$b^2 = 2k^2.$$

Thus b^2 is even, so b is even.

We have shown both a and b are even, so they share a factor of 2. This contradicts the assumption that a/b was in lowest terms. Therefore $\sqrt{2}$ is irrational. □

8.2 The historical shock

The discovery of irrational magnitudes is often associated with the Pythagorean tradition. Whether the old stories are embellished or not, the mathematical point is real: geometry produced lengths that arithmetic, if limited to ratios of whole numbers, could not contain.

The diagonal of a unit square has length $\sqrt{2}$. That length is perfectly visible geometrically and impossible to express as a ratio of integers. Geometry forced number to grow.

8.3 The structure of the proof

The proof has several moving parts:

1. Assume rationality.
2. Choose a fraction in lowest terms.
3. Derive that numerator and denominator are both even.
4. Contradict lowest terms.

The contradiction is not "something feels wrong." It is a direct collision between "lowest terms" and "both divisible by 2."

Common trap

In a contradiction proof, name the impossible thing. A vague contradiction is not enough. Say exactly which statement conflicts with which earlier assumption.

8.4 Exercises

1. Prove that $\sqrt{3}$ is irrational. You may use: if $3 \mid n^2$, then $3 \mid n$.
2. Prove that $\sqrt{5}$ is irrational using the analogous divisibility fact.
3. Prove there is no rational number r such that $r^2 = 2$.
4. Explain why the lowest-terms assumption matters.

9 Examples, Non-Examples, and Counterexamples

Examples are not afterthoughts. They are the laboratory where definitions become real.

9.1 Example versus proof

An example can prove an existential statement:

$$\exists n \in \mathbb{Z} \text{ such that } n^2 = 25.$$

Take $n = 5$.

But an example cannot prove a universal statement:

$$\forall n \in \mathbb{Z},\ n^2 \geq 0.$$

Checking $n = 5$ helps you understand, but it does not prove the statement.

9.2 Counterexamples

A counterexample disproves a universal claim.

Claim:

$$\forall x, y \in \mathbb{R},\ x^2 = y^2 \Rightarrow x = y.$$

Counterexample: $x = 1$, $y = -1$. Then $x^2 = y^2$, but $x \neq y$.

9.3 Near misses

A near miss almost satisfies a definition but fails one condition. Near misses are powerful because they reveal why the missing condition matters.

Example: A function can be injective but not surjective.

Let $f : \mathbb{N} \to \mathbb{N}$ by $f(n) = n + 1$. It is injective: different inputs give different outputs. It is not surjective if $\mathbb{N} = \{1, 2, 3, \ldots\}$, because 1 is never hit.

Example: A function can be surjective but not injective.

Let $g : \{1, 2\} \to \{a\}$ by $g(1) = a$ and $g(2) = a$. It hits every element of the codomain, but merges two inputs.

9.4 The counterexample habit

Whenever you learn a theorem, remove one assumption and try to break it.

Theorem:

> A continuous function on a closed bounded interval attains a maximum.

Remove closed:

$$f(x) = x \text{ on } (0, 1)$$

has no maximum.

Remove bounded:

$$f(x) = x \text{ on } [0, \infty)$$

has no maximum.

Remove continuous: a function on $[0, 1]$ that jumps upward near a missing value can fail to attain a supremum.

History and connection

Much of real analysis was shaped by counterexamples. Mathematicians learned that intuitive claims about functions could fail in strange ways. The response was not to abandon intuition, but to sharpen definitions.

9.5 Exercises

For each false statement, find a counterexample. Then explain which assumption fails.

1. Every injective function is surjective.
2. Every surjective function is injective.
3. Every bounded sequence converges.
4. Every continuous function is differentiable.
5. If $A \cup B = A \cup C$, then $B = C$.
6. If $ab = 0$, then $a = 0$ and $b = 0$.

10 Sets: The Grammar of Modern Mathematics

A set is a collection of objects. The objects are called elements.

$$x \in A$$

means x is an element of A.

$$x \notin A$$

means x is not an element of A.

This seems almost too simple. But sets became the basic grammar of modern mathematics because almost anything can be collected: numbers, points, functions, matrices, solutions, open sets, symmetries, curves, ideals.

10.1 Basic set operations

Definition 6. The union of A and B is

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

The "or" is inclusive. If x belongs to both, it belongs to the union.

Definition 7. The intersection of A and B is

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

Definition 8. The difference of A and B is

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

Definition 9. The power set of A, written $\mathcal{P}(A)$, is the set of all subsets of A.

If $A = \{1, 2\}$, then

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

10.2 Subsets

Definition 10. $A \subseteq B$ means every element of A is an element of B.

To prove $A \subseteq B$, use the universal structure:

> Let $x \in A$ be arbitrary. Show $x \in B$.

To disprove $A \subseteq B$, find one $x \in A$ with $x \notin B$.

10.3 Set equality

To prove $A = B$, prove:

$$A \subseteq B \quad \text{and} \quad B \subseteq A.$$

This is often called proving both inclusions.

Intuition

Set equality is mutual containment. Nothing on the left is missing from the right, and nothing on the right is extra.

10.4 Historical connection

Cantor's set theory made infinity into an object of study. Sets allowed mathematicians to compare the sizes of infinite collections, define functions carefully, and eventually build the foundations for analysis and topology. Set theory also produced paradoxes when handled too freely, which pushed mathematics toward more careful axioms.

11 Set Proofs: Element Chasing in Detail

Set proofs often feel mechanical at first. That is not bad. You are learning grammar. Later, the grammar becomes invisible enough that you can think with it.

11.1 A simple inclusion

Theorem 7. *For all sets A and B, $A \cap B \subseteq A$.*

Proof. Let $x \in A \cap B$ be arbitrary. By definition of intersection, $x \in A$ and $x \in B$. In particular, $x \in A$. Therefore $A \cap B \subseteq A$. □

11.2 A set equality

Theorem 8. *For all sets A and B,*

$$A \setminus B = A \cap B^c.$$

Proof. Let x be arbitrary. Then

$$x \in A \setminus B$$

if and only if $x \in A$ and $x \notin B$. The condition $x \notin B$ is equivalent to $x \in B^c$. Thus $x \in A$ and $x \in B^c$, which is equivalent to $x \in A \cap B^c$. Therefore $A \setminus B = A \cap B^c$. □

11.3 A distributive law

Theorem 9. *For all sets A, B, C,*

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

Proof. We prove both inclusions.

First let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$. Since $x \in B \cup C$, either $x \in B$ or $x \in C$.

If $x \in B$, then $x \in A \cap B$, so $x \in (A \cap B) \cup (A \cap C)$.

If $x \in C$, then $x \in A \cap C$, so $x \in (A \cap B) \cup (A \cap C)$.

Thus $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$.

Now let $x \in (A \cap B) \cup (A \cap C)$. Then either $x \in A \cap B$ or $x \in A \cap C$.

If $x \in A \cap B$, then $x \in A$ and $x \in B$. Since $x \in B$, we have $x \in B \cup C$. Therefore $x \in A \cap (B \cup C)$.

If $x \in A \cap C$, then $x \in A$ and $x \in C$. Since $x \in C$, we have $x \in B \cup C$. Therefore $x \in A \cap (B \cup C)$.

Thus $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$. Therefore the sets are equal. □

11.4 Exercises

Prove each by element chasing.

1. $A \subseteq A \cup B$.
2. $A \cap (B \setminus C) = (A \cap B) \setminus C$.
3. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
4. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.
5. If $A \subseteq B$, then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

12 Functions: Structured Promises, Not Just Formulas

A function is one of the central objects in all of mathematics.

Definition 11. A function $f : A \to B$ assigns to every element of A exactly one element of B.

The set A is the domain. The set B is the codomain.

12.1 Exactly one

The phrase "exactly one" contains two promises:

- every input in A gets an output,
- no input gets two different outputs.

A relation that sends one input to two outputs is not a function. A rule that fails to define an output for some input in the domain is not a function from that domain.

12.2 Formula is not enough

The formula x^2 can define different functions:

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^2.$$

$$g : \mathbb{R} \to [0, \infty), \quad g(x) = x^2.$$

$$h : [0, \infty) \to [0, \infty), \quad h(x) = x^2.$$

The formula is the same, but the functions differ because domain and codomain are part of the data.

12.3 Image

Definition 12. The image of $f : A \to B$ is

$$\mathrm{im}(f) = \{f(a) : a \in A\}.$$

The image is the part of the codomain actually hit.

For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, the codomain is $\mathbb{R}$, but the image is $[0, \infty)$.

12.4 Why software engineers should care

The type signature matters. A function's declared output type may be larger than the values it actually returns. Surjectivity asks whether the declared output space is fully covered. Injectivity asks whether identity of inputs can be recovered from outputs.

13 Injective, Surjective, Bijective

13.1 Injective

Definition 13. A function $f : A \to B$ is injective if for all $a_1, a_2 \in A$,

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

Injective means no merging. If two outputs are equal, the inputs must have been equal.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = 2x + 1$$

is injective. If $2a + 1 = 2b + 1$, then $a = b$.

Non-example:

$$g : \mathbb{R} \to \mathbb{R}, \quad g(x) = x^2$$

is not injective, since $g(1) = g(-1)$ but $1 \neq -1$.

13.2 Surjective

Definition 14. A function $f : A \to B$ is surjective if for every $b \in B$, there exists $a \in A$ such that $f(a) = b$.

Surjective means full coverage of the codomain.

Example:

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^3$$

is surjective. Given $y \in \mathbb{R}$, choose $x = \sqrt[3]{y}$.

Non-example:

$$g : \mathbb{R} \to \mathbb{R}, \quad g(x) = x^2$$

is not surjective, since no real x satisfies $x^2 = -1$.

13.3 Bijective

Definition 15. A function is bijective if it is both injective and surjective.

A bijection pairs the elements of the domain with the elements of the codomain perfectly: no merging and no missing.

13.4 Proof patterns

To prove injective:

> Suppose $f(a_1) = f(a_2)$. Show $a_1 = a_2$.

To prove surjective:

> Let $b \in B$ be arbitrary. Find $a \in A$ such that $f(a) = b$.

Proof lab

Prove that $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 5x - 7$ is bijective. For injective, start with $f(a) = f(b)$. For surjective, given y, solve $5x - 7 = y$ for x.

14 Images and Inverse Images

Images and inverse images are where functions meet sets.

14.1 Image of a set

If $f : A \to B$ and $S \subseteq A$, then

$$f(S) = \{f(s) : s \in S\}.$$

This is the set of outputs produced by inputs from S.

14.2 Inverse image of a set

If $T \subseteq B$, then

$$f^{-1}(T) = \{a \in A : f(a) \in T\}.$$

This is the set of inputs whose outputs land in T.

Common trap

The notation $f^{-1}(T)$ does not require f to have an inverse function. It is an inverse image of a set, not necessarily an inverse function.

14.3 Worked proof

Theorem 10. *For $f : A \to B$ and $S, T \subseteq B$,*

$$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T).$$

Proof. Let $a \in A$. Then

$$a \in f^{-1}(S \cap T)$$

if and only if $f(a) \in S \cap T$. This holds if and only if $f(a) \in S$ and $f(a) \in T$. That is equivalent to $a \in f^{-1}(S)$ and $a \in f^{-1}(T)$, which means $a \in f^{-1}(S) \cap f^{-1}(T)$. Therefore the sets are equal. □

14.4 A place where image behaves worse

It is always true that

$$f(S \cap T) \subseteq f(S) \cap f(T).$$

But equality can fail. Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$, let $S = \{-1\}$ and $T = \{1\}$. Then $S \cap T = \emptyset$, so $f(S \cap T) = \emptyset$. But

$$f(S) = \{1\} = f(T),$$

so

$$f(S) \cap f(T) = \{1\}.$$

The function merged -1 and 1. This is exactly the kind of problem injectivity controls.

14.5 Connection to topology

Topology defines continuity using inverse images. A function between topological spaces is continuous if the inverse image of every open set is open. This is why inverse images deserve attention now. They become a central language later.

15 Relations and Equivalence

A relation is a way of saying which objects are connected.

Examples:

- equality,
- less than,
- divisibility,
- congruence modulo n,
- "has the same birthday as,"
- "is connected by a path to."

15.1 Equivalence relations

Definition 16. A relation $\sim$ on a set A is an equivalence relation if it is reflexive, symmetric, and transitive.

Reflexive:

$$a \sim a.$$

Symmetric:

$$a \sim b \Rightarrow b \sim a.$$

Transitive:

$$a \sim b \text{ and } b \sim c \Rightarrow a \sim c.$$

15.2 What equivalence means

An equivalence relation says: for the purpose at hand, these objects count as the same.

Fractions are the everyday example:

$$\frac{1}{2}, \frac{2}{4}, \frac{3}{6}$$

are different symbols but represent the same rational number.

Congruence modulo 5 says:

$$2, 7, 12, 17, \ldots$$

count as the same remainder class.

15.3 Worked proof

Theorem 11. *Congruence modulo n is an equivalence relation on $\mathbb{Z}$.*

Proof. Define $a \sim b$ if $n \mid (a-b)$.

Reflexive: For any $a \in \mathbb{Z}$, $a - a = 0$, and $n \mid 0$, so $a \sim a$.

Symmetric: Suppose $a \sim b$. Then $n \mid (a-b)$, so $a - b = nk$ for some $k \in \mathbb{Z}$. Then $b - a = -nk = n(-k)$, so $n \mid (b-a)$. Therefore $b \sim a$.

Transitive: Suppose $a \sim b$ and $b \sim c$. Then $a - b = nm$ and $b - c = n\ell$ for integers m, ℓ. Adding gives

$$a - c = (a-b) + (b-c) = nm + n\ell = n(m+\ell).$$

Thus $n \mid (a-c)$, so $a \sim c$. □

15.4 Quotients

An equivalence relation partitions a set into equivalence classes. The set of equivalence classes is a quotient.

Quotients are everywhere later:

- modular arithmetic,
- quotient groups,
- quotient rings,
- projective space,
- gluing edges of polygons to make surfaces,
- identifying points in topology.

Walking question

Ask: what differences are being ignored? A quotient is often a mathematical way to say "we no longer care about this distinction."

16 Orders: Another Kind of Relation

Not every important relation is equivalence. Some relations compare rather than identify.

Definition 17. A relation $\leq$ on a set A is a partial order if it is reflexive, antisymmetric, and transitive.

Reflexive:

$$a \leq a.$$

Antisymmetric:

$$a \leq b \text{ and } b \leq a \Rightarrow a = b.$$

Transitive:

$$a \leq b \text{ and } b \leq c \Rightarrow a \leq c.$$

16.1 Examples

The usual order on $\mathbb{R}$ is a partial order.

Subset inclusion $\subseteq$ is a partial order on $\mathcal{P}(A)$.

Divisibility is a partial order on positive integers: $a \preceq b$ if $a \mid b$.

16.2 Partial versus total

In a total order, any two elements can be compared. In a partial order, some pairs may be incomparable.

For subsets of $\{1, 2, 3\}$, the sets $\{1\}$ and $\{2\}$ are incomparable under $\subseteq$: neither is contained in the other.

16.3 Why orders matter

Orders appear in lattice theory, topology, logic, domain theory in computer science, divisibility, and algebraic geometry. Inclusion of ideals is a partial order. Open sets are ordered by inclusion. Dependencies in software form a partial order when cycles are absent.

16.4 Exercises

1. Prove that $\subseteq$ is a partial order on $\mathcal{P}(A)$.

2. Prove that divisibility is a partial order on $\mathbb{N}$.
3. Give an example of a relation that is reflexive and transitive but not antisymmetric.
4. Give an example of a partial order that is not total.

17 Induction: Infinite Claims, Finite Proofs

Induction is a proof machine for statements indexed by natural numbers.

Theorem 12 (Induction)**.** *Suppose $P(n)$ is a statement for each integer $n \geq 1$. If $P(1)$ is true and $P(k) \Rightarrow P(k+1)$ for every $k \geq 1$, then $P(n)$ is true for all $n \geq 1$.*

17.1 The domino picture

If the first domino falls, and each domino knocks down the next, then every domino falls.

The danger is that the picture can make induction feel less precise than it is. The induction step must prove:

$$P(k) \Rightarrow P(k+1)$$

for an arbitrary k.

17.2 Worked proof

Theorem 13. *For all $n \geq 1$,*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Base case: For $n = 1$, the left side is 1 and the right side is $\frac{1(2)}{2} = 1$, so the statement is true.

Induction step: Assume for some $k \geq 1$ that

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

We must prove

$$1 + 2 + \cdots + k + (k+1) = \frac{(k+1)(k+2)}{2}.$$

Using the induction hypothesis,

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1).$$

Now

$$\frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}.$$

So the statement holds for $k+1$. By induction, it holds for every $n \geq 1$. □

17.3 Common induction mistakes

- Forgetting the base case.
- Assuming $P(k+1)$ instead of $P(k)$.
- Proving only one example of the induction step.
- Not saying where the induction hypothesis is used.

17.4 Exercises

Prove by induction.

1. $1 + 3 + 5 + \cdots + (2n-1) = n^2$.
2. $2^n \geq n + 1$ for all $n \geq 1$.
3. $3 \mid (4^n - 1)$ for all $n \geq 1$.
4. $n! \geq 2^{n-1}$ for all $n \geq 1$.

18 Strong Induction and Recursive Thinking

Strong induction lets the induction step use all previous cases, not just the immediately previous one.

Theorem 14 (Strong induction)**.** *Suppose $P(n)$ is a statement for integers $n \geq n_0$. If $P(n_0)$ is true and for every $k \geq n_0$, the truth of $P(n_0), P(n_0+1), \ldots, P(k)$ implies $P(k+1)$, then $P(n)$ is true for all $n \geq n_0$.*

18.1 Why it is not stronger in truth, but stronger in convenience

Ordinary induction and strong induction are logically equivalent. But strong induction often matches the problem better. If an object can break into smaller pieces of many possible sizes, you want to assume all smaller cases are known.

18.2 Prime factorization

Theorem 15. *Every integer $n \geq 2$ is a product of primes.*

Proof. We use strong induction.

Base case: 2 is prime, so 2 is a product of primes.

Induction step: Assume every integer m with $2 \leq m \leq k$ is a product of primes. We prove $k+1$ is a product of primes.

If $k+1$ is prime, then it is already a product of primes.

If $k+1$ is composite, then $k+1 = ab$ where $2 \leq a \leq k$ and $2 \leq b \leq k$. By the induction hypothesis, both a and b are products of primes. Therefore $ab = k+1$ is a product of primes.

Thus every integer $n \geq 2$ is a product of primes. □

18.3 Connection to algorithms

Strong induction resembles recursive correctness proofs. If an algorithm solves a problem by reducing it to smaller subproblems, you prove correctness by assuming the algorithm works on all smaller inputs and showing it works on the current one.

18.4 Exercises

1. Prove every integer $n \geq 2$ is either prime or divisible by a prime.

2. Prove every amount of postage $n \geq 12$ can be made using only 4-cent and 5-cent stamps.
3. Prove the Fibonacci identity $F_{n+2} = F_{n+1} + F_n$ is built into the recursive definition, then prove a nontrivial Fibonacci sum identity.

19 Countability and the First Shock of Infinity

Infinity is not one thing.

Definition 18. A set is countably infinite if its elements can be put in one-to-one correspondence with $\mathbb{N}$.

Informally: you can list the elements in a sequence so that every element eventually appears.

19.1 The even numbers are countable

The map

$$f : \mathbb{N} \to 2\mathbb{N}, \quad f(n) = 2n$$

is a bijection. So the positive even integers have the same cardinality as the positive integers.

This feels strange at first because the evens are a proper subset of the naturals. Infinite sets do that.

19.2 The integers are countable

List:

$$0, 1, -1, 2, -2, 3, -3, \ldots$$

Every integer appears.

19.3 The rationals are countable

Rationals feel much larger because they are dense: between any two rationals there is another rational. But density is not cardinality.

Arrange fractions p/q in a grid with $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. Sweep through the grid by increasing $|p| + q$, skipping duplicates. Every rational appears after finitely many steps.

19.4 The reals are uncountable

Cantor's diagonal argument shows that no list of real numbers in $(0, 1)$ can be complete.

Suppose someone gives a list:

$$r_1, r_2, r_3, \ldots$$

Write their decimal expansions. Construct a new number s whose first digit differs from the first digit of r_1, whose second digit differs from the second digit of r_2, and so on. Then s differs from every r_i in at least one decimal place. So s is not on the list.

Therefore the reals in $(0, 1)$ are uncountable.

19.5 Why this matters later

Analysis lives in uncountable spaces. Topology studies infinite collections of open sets. Algebraic geometry studies solution sets that can be infinite but controlled by finitely many equations. Cantor's work made it possible to talk about infinity without hand waving.

20 Real Analysis Preview: Epsilon as a Contract

Calculus says a sequence approaches a limit. Analysis asks what that sentence means.

Definition 19. A sequence (a_n) converges to L if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|a_n - L| < \varepsilon.$$

20.1 Plain English

No matter how small a tolerance ε someone demands, you can go far enough into the sequence that every later term stays within that tolerance of L.

20.2 Who chooses what?

The challenger chooses $\varepsilon > 0$. You choose N. Then the challenger may choose any $n \geq N$, and you must be safe.

20.3 Worked proof

Theorem 16.

$$\lim_{n\to\infty} \frac{1}{n} = 0.$$

Proof. Let $\varepsilon > 0$ be arbitrary. Choose $N \in \mathbb{N}$ such that $N > 1/\varepsilon$. If $n \geq N$, then

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Therefore $1/n \to 0$. □

20.4 Continuity

Definition 20. A function $f : \mathbb{R} \to \mathbb{R}$ is continuous at c if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x \in \mathbb{R}$, if $|x - c| < \delta$, then $|f(x) - f(c)| < \varepsilon$.

Theorem 17. *$f(x) = 3x + 2$ is continuous at every $c \in \mathbb{R}$.*

Proof. Let $c \in \mathbb{R}$ and $\varepsilon > 0$. Choose $\delta = \varepsilon/3$. If $|x - c| < \delta$, then

$$|f(x) - f(c)| = |(3x + 2) - (3c + 2)| = 3|x - c| < 3\delta = \varepsilon.$$

Thus f is continuous at c. □

20.5 Why analysis is proof training

Analysis is quantifier training under pressure. You must track dependencies carefully: N depends on ε; δ depends on ε and often on the point c; variables must be chosen in the correct order.

21 Abstract Algebra Preview: Structure and Symmetry

Abstract algebra begins when you stop caring what the objects are made of and start caring how they combine.

Definition 21. A group is a set G with an operation $\cdot$ satisfying closure, associativity, identity, and inverses.

21.1 Examples

- $\mathbb{Z}$ under addition.
- Nonzero real numbers under multiplication.
- Symmetries of a square under composition.
- Invertible $n \times n$ matrices under multiplication.

21.2 Identity is unique

Theorem 18. *The identity element of a group is unique.*

Proof. Suppose e and e' are both identity elements. Since e is an identity, $ee' = e'$. Since e' is an identity, $ee' = e$. Therefore $e = e'$. Hence the identity is unique. □

21.3 Inverses are unique

Theorem 19. *Every element of a group has a unique inverse.*

Proof. Let $a \in G$. Suppose b and c are both inverses of a. Then $ab = e$ and $ca = e$. Now

$$b = eb = (ca)b = c(ab) = ce = c.$$

Therefore the inverse of a is unique. □

21.4 Why algebra feels abstract

The proof did not use numbers. It used only the group laws. That is the power. Any result proved from group axioms applies to every group.

History and connection

Modern algebra grew from attempts to solve polynomial equations and understand symmetry. Galois connected the solvability of equations to groups of symmetries. Noether later helped shift the subject toward structural thinking, where operations and maps became central.

22 Topology Preview: Continuity Without Rulers

Topology studies spaces by their open sets and continuous maps.

Definition 22. A topology on a set X is a collection $\mathcal{T}$ of subsets of X such that:

1. $\emptyset$ and X are in $\mathcal{T}$,
2. arbitrary unions of sets in $\mathcal{T}$ are in $\mathcal{T}$,
3. finite intersections of sets in $\mathcal{T}$ are in $\mathcal{T}$.

The sets in $\mathcal{T}$ are called open sets.

22.1 Why these axioms?

Open sets are meant to capture neighborhoods, regions where points have room. Arbitrary unions of open regions should be open because if a point lies in one of the regions, it has room there. Finite intersections should be open because a point lying in finitely many open regions has room small enough to stay inside all of them.

Infinite intersections are not required to be open. For example:

$$\bigcap_{n=1}^{\infty}(-1/n, 1/n) = \{0\},$$

which is not open in $\mathbb{R}$.

22.2 Continuity

Definition 23. If X and Y are topological spaces, a function $f : X \to Y$ is continuous if for every open set $U \subseteq Y$, the inverse image $f^{-1}(U)$ is open in X.

This is why inverse images mattered earlier. Topology takes the epsilon-delta idea and expresses it without distances.

22.3 Connection to geometry

Topology ignores exact lengths and angles. It cares about connectedness, holes, compactness, and continuity. Differential geometry later puts smooth structure and calculus back on top of topological spaces. Algebraic geometry gives spaces built from polynomial equations a topology of their own.

23 Differential Geometry Preview: Calculus on Curved Worlds

Differential geometry asks: how do we do calculus on spaces that are not flat?

A surface like a sphere is not globally the plane, but near any small patch it resembles the plane. This is the core idea behind manifolds.

Definition 24. Informally, an n-dimensional manifold is a space that locally looks like $\mathbb{R}^n$.

23.1 Charts

A chart assigns coordinates to a small region of a manifold. Think of a map of Earth. No flat map captures the whole globe perfectly, but local maps work.

23.2 Tangent spaces

At each point of a smooth manifold, there is a tangent space: a linear approximation to the manifold near that point.

This connects to linear algebra. Derivatives become linear maps between tangent spaces.

23.3 Physical connection

Differential geometry is the language of curvature. General relativity uses curved spacetime. Robotics and computer graphics use rotations, configurations, and smooth spaces. Optimization on manifolds appears in machine learning and data analysis.

23.4 Why proof foundations matter

Manifolds require comfort with functions, open sets, local-to-global reasoning, equivalence of coordinate descriptions, and linear maps. The early proof habits are not separate from geometry; they are the scaffolding that lets geometry become precise.

24 Algebraic Geometry Preview: Equations as Shapes

Algebraic geometry studies solution sets of polynomial equations using algebra.

24.1 First examples

The equation

$$x^2 + y^2 = 1$$

defines a circle.

The equation

$$y^2 = x^3 - x$$

defines an elliptic curve.

A system of equations defines the set of points satisfying all equations simultaneously.

Definition 25. For polynomials $f_1, \ldots, f_m \in k[x_1, \ldots, x_n]$ over a field k, the affine variety they define is

$$V(f_1, \ldots, f_m) = \{a \in k^n : f_i(a) = 0 \text{ for all } i\}.$$

24.2 Ideals

Definition 26. An ideal I in a ring R is a nonempty subset such that:

1. if $a, b \in I$, then $a + b \in I$,
2. if $r \in R$ and $a \in I$, then $ra \in I$.

Ideals encode equations. Geometry becomes algebra when you study the polynomials that vanish on a shape.

24.3 Why the early topics matter

Algebraic geometry uses:

- sets to describe solution collections,
- functions to describe maps between spaces,
- rings and ideals from algebra,
- topology to describe geometric behavior,

- quotients to identify algebraic structures,
- proof maturity to survive abstraction.

History and connection

Algebraic geometry began with concrete curves and surfaces. Over time it absorbed commutative algebra, topology, category theory, and number theory. Grothendieck's revolution was not making the subject obscure for fun; it created a language powerful enough to see many older phenomena as parts of one architecture.

25 Proof Debugging: Turning Fog Into Work

When a proof fails, do not panic and do not start over immediately. Debug it.

25.1 The proof debugger

1. State the claim exactly.
2. Identify all variables and domains.
3. Circle the quantifiers.
4. Separate hypotheses from conclusion.
5. Expand definitions in the conclusion.
6. Expand definitions in the hypotheses.
7. Choose a strategy.
8. Write a skeleton.
9. Fill in one line at a time.
10. Check every line: why does it follow?

25.2 Gap types

Gap	Symptom	Repair
Quantifier gap	Object used before being introduced	Introduce it with domain and role
Definition gap	Term used vaguely	Replace term with definition
Existence gap	Object claimed without witness	Construct or cite existence
Implication gap	A line jumps too far	Add intermediate justification
Case gap	Not all possibilities handled	List cases explicitly
Converse error	Direction reversed	Check hypothesis and conclusion

25.3 Bad proof example

Claim: If n^2 is even, then n is even.

Bad proof:

> Since n^2 is even, $n^2 = 2k$. Taking square roots gives $n = \sqrt{2k}$, so n is even.

The broken line is the last one. Being $\sqrt{2k}$ does not mean being twice an integer. Repair by using the contrapositive.

25.4 What good feedback sounds like

Bad feedback:

> This proof is wrong.

Useful feedback:

> The first unsupported step is line 3. You need to show $\sqrt{2k}$ has the form $2m$ for some integer m, but that is false in general.

26 A Serious Velleman Readiness Plan

You are ready to read Velleman well when the following tasks feel possible, even if not effortless.

26.1 Core readiness

1. Translate English statements into quantifiers.
2. Negate nested quantified statements.
3. Prove direct claims from definitions.
4. Use contrapositive when the conclusion has a useful negation.
5. Use contradiction with a named contradiction.
6. Prove set equality by two inclusions.
7. Prove injectivity and surjectivity from definitions.
8. Work with inverse images of sets.
9. Prove a relation is an equivalence relation.
10. Write induction proofs with base case, hypothesis, and step.

26.2 Four-week deep launch

Week	Theme	Daily work	Milestone
1	Sentences and quantifiers	Translate, negate, identify variable roles	Quantifier order feels meaningful
2	Definitions and proof moves	Parity, divisibility, rationality, contradiction	You can write clean one-page proofs
3	Sets and functions	Element chasing, image, inverse image, injective/surjective	You can prove function claims from definitions
4	Relations and induction	Equivalence, orders, induction, countability preview	You can begin Velleman without fog dominating

26.3 Portfolio

Write polished proofs of:

1. sum of evens is even,

2. product of odds is odd,
3. divisibility is transitive,
4. if n^2 is even then n is even,
5. $\sqrt{2}$ is irrational,
6. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$,
7. $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$,
8. composition of injective functions is injective,
9. congruence modulo n is an equivalence relation,
10. $1 + 2 + \cdots + n = n(n + 1)/2$.

27 Second Pass: Logic as a Language You Can Speak

The first pass taught the parts. This second pass is where the parts become usable. We revisit the same concepts, but from different angles. This is not repetition for padding. It is repetition because the same idea looks different when you approach it as a reader, a writer, a debugger, and a future analyst or algebraist.

27.1 The reader's view

When you read a mathematical sentence, your first job is to identify its grammar. Consider:

$$\forall n \in \mathbb{Z},\ n^2 + n \text{ is even.}$$

The outer shape is universal. It says every integer has a property. A proof must begin with an arbitrary integer:

> Let $n \in \mathbb{Z}$ be arbitrary.

The property is "$n^2 + n$ is even." By definition, proving this means finding an integer k such that

$$n^2 + n = 2k.$$

Now the proof has a target. We are not just trying to say something plausible about $n^2 + n$. We are trying to put it into the shape $2k$.

27.2 The writer's view

A proof might go:

Proof. Let $n \in \mathbb{Z}$ be arbitrary. Then

$$n^2 + n = n(n + 1).$$

One of n and $n + 1$ is even, so $n(n + 1)$ is even. Therefore $n^2 + n$ is even. □

This proof is short, but it hides a lemma: one of two consecutive integers is even. For a beginner portfolio, you might expand it:

Proof. Let $n \in \mathbb{Z}$ be arbitrary. If n is even, then $n = 2m$ for some $m \in \mathbb{Z}$, so

$$n^2 + n = n(n + 1) = 2m(n + 1),$$

which is even. If n is odd, then $n = 2m + 1$ for some $m \in \mathbb{Z}$, so

$$n + 1 = 2m + 2 = 2(m + 1).$$

Thus

$$n(n + 1) = n \cdot 2(m + 1) = 2(n(m + 1)),$$

which is even. In both cases, $n^2 + n$ is even. □

The longer proof is not more advanced. It is more explicit. At the beginning, explicit is good.

27.3 The debugger's view

Bad proof:

> $n^2 + n$ is even because it has two terms and one is odd and one is even.

Diagnosis:

- "two terms" has no direct connection to evenness;
- n^2 and n are not always one odd and one even;
- the proof never uses the definition of even.

Repair:

- factor $n^2 + n = n(n + 1)$;
- use the fact that consecutive integers have opposite parity;
- close the definition of even.

27.4 The later-math view

The expression $n(n + 1)$ being even is a divisibility statement. It says:

$$2 \mid n(n + 1).$$

In number theory and abstract algebra, divisibility becomes a structural idea. In modular arithmetic, the same proof can be written:

$$n \equiv 0 \pmod 2 \quad \text{or} \quad n \equiv 1 \pmod 2.$$

If $n \equiv 0$, then $n(n + 1) \equiv 0$. If $n \equiv 1$, then $n + 1 \equiv 0$, so again $n(n + 1) \equiv 0$. The same idea has moved into the language of congruence.

Proof lab

Write three proofs that $n^2 + n$ is even: one by cases, one using modular arithmetic, and one using the phrase "consecutive integers." Then decide which proof is best for a beginner and which is best for a reader who already knows congruences.

28 Second Pass: Quantifier Order in Real Problems

Quantifier order is easy to understand in toy examples and easy to mishandle in serious mathematics. The way to improve is to translate each statement into a game of choices.

28.1 Uniform versus pointwise

Compare:

$$\forall x \in A, \exists \delta > 0,\ P(x, \delta)$$

with

$$\exists \delta > 0, \forall x \in A,\ P(x, \delta).$$

The first allows δ to depend on x. The second requires one δ that works for all x.

This distinction appears everywhere.

28.2 Continuity versus uniform continuity

Continuity at each point says, roughly:

$$\forall x, \forall \varepsilon > 0, \exists \delta > 0, \forall y,\ \text{if } |y - x| < \delta \text{ then } |f(y) - f(x)| < \varepsilon.$$

The δ may depend on both ε and the point x.

Uniform continuity says:

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x, \forall y,\ \text{if } |y - x| < \delta \text{ then } |f(y) - f(x)| < \varepsilon.$$

Now δ depends only on ε, not on the location x.

28.3 Physical picture

Imagine a road with changing steepness. Ordinary continuity says: at every location, if you zoom in enough, small horizontal movement produces small vertical movement. But the amount of zoom may depend on where you stand.

Uniform continuity says: one zoom level works across the whole road.

On a closed bounded interval, continuous functions are uniformly continuous. That theorem is a jewel of real analysis because it converts local control into global control. Compactness is the hidden machine behind that conversion.

28.4 Software picture

Suppose a function has a stability guarantee:

> For every input, there exists a tolerance setting that keeps output error below 0.01.

That is pointwise. It may require retuning for every input.

Now compare:

> There exists one tolerance setting that works for every input.

That is uniform. It is much stronger and much more useful operationally.

28.5 Exercises

For each pair, explain which statement is stronger.

1. $\forall x \in A, \exists y \in B,\ R(x, y)$ versus $\exists y \in B, \forall x \in A,\ R(x, y)$.
2. $\forall p \in X, \exists r > 0,\ B_r(p) \subseteq U$ versus $\exists r > 0, \forall p \in X,\ B_r(p) \subseteq U$.
3. For every bug there exists a test that catches it versus there exists one test that catches every bug.
4. For every city there is a road to the capital versus there is one road that goes from every city to the capital.

Walking question

When walking, invent three everyday statements of the form "for every, there exists." Then reverse the quantifiers and ask whether the new statement is still true.

29 Second Pass: Definitions Under a Microscope

Definitions are compact. That is why they are useful and why they are dangerous. A definition may compress a page of intuition into one line. If you read only the line, you may miss the intuition. If you keep only the intuition, you may lose the rigor.

29.1 Case study: injective

Formal definition:

$$f : A \to B \text{ is injective if } \forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

Plain version: equal outputs force equal inputs.

Visual version: no two distinct inputs point to the same output.

Information version: the output retains enough information to recover the input.

Proof version: to prove injective, assume $f(a_1) = f(a_2)$ and show $a_1 = a_2$.

Counterexample version: to disprove injective, find $a_1 \neq a_2$ with $f(a_1) = f(a_2)$.

29.2 Why the official definition uses equal outputs

Beginners often define injective as:

> Different inputs give different outputs.

That is true:

$$a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2).$$

But the standard definition uses the contrapositive:

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2.$$

Why? Because in algebraic proofs, it is often easier to start with equality of outputs and manipulate equations until equality of inputs appears.

29.3 Case study: open set

Metric definition:

$$U \subseteq X \text{ is open if } \forall x \in U, \exists r > 0,\ B_r(x) \subseteq U.$$

Plain version: every point of U has wiggle room inside U.

Physical version: if you stand at any point in the region, you can move a little in any direction without leaving the region.

Proof version: to prove open, start with arbitrary $x \in U$ and produce a radius $r > 0$.

Counterexample version: to prove not open, find a point in the set with no positive-radius ball contained in the set.

29.4 The boundary tells the story

The interval $(0, 1)$ is open in $\mathbb{R}$. If $x \in (0, 1)$, choose

$$r = \frac{1}{2}\min(x, 1 - x).$$

Then the ball $(x - r, x + r)$ stays inside $(0, 1)$.

The interval $[0, 1]$ is not open in $\mathbb{R}$ because at $x = 0$, every ball around 0 contains negative numbers outside $[0, 1]$.

29.5 Later connection

Open sets become the basic language of topology. The same definition pattern appears in manifolds: local neighborhoods look like open subsets of $\mathbb{R}^n$. Algebraic geometry has its own topologies, where open sets may behave very differently from metric open sets.

Proof lab

For each definition you meet this week, write six versions: formal, plain, visual, proof, counterexample, and later connection.

30 Second Pass: Proofs as Programs With Human Readers

A proof resembles a program in that each line must be justified. But a proof is also unlike a program because its compiler is a human reader with mathematical expectations.

30.1 Types

Every object has a type:

- $n \in \mathbb{Z}$,
- $x \in \mathbb{R}$,
- $A \subseteq X$,
- $f : A \to B$,
- G is a group,
- U is open.

Type errors in proofs are common. You cannot write $x \in f$ if f is a function and x is not a function element. You cannot ask whether $A < B$ if A and B are sets without an order relation.

30.2 Preconditions and postconditions

In a theorem:

> If $f : A \to B$ is injective and $g : B \to C$ is injective, then $g \circ f : A \to C$ is injective.

The preconditions are:

- f is a function from A to B,
- g is a function from B to C,
- f is injective,
- g is injective.

The postcondition is:

- $g \circ f$ is injective.

Proof writing becomes easier when you treat the conclusion as a spec. What exactly must be shown for $g \circ f$ to satisfy the injective spec?

30.3 Invariants

In induction, the statement $P(n)$ is like an invariant over natural numbers. The base case initializes it. The induction step preserves it from k to $k + 1$.

This analogy is not perfect, but it helps. A loop invariant must hold before and after each iteration. An induction statement must propagate from one case to the next.

30.4 Readable proof style

A proof should not be cryptic. Good proof prose says what is happening:

> We prove both inclusions.
>
> Let $x \in A$ be arbitrary.
>
> By injectivity of g, we get $f(a_1) = f(a_2)$.

These phrases are not filler. They orient the reader.

30.5 Exercise

Take a proof you wrote. Annotate it like code:

- mark variable introductions,
- mark definitions opened,
- mark definitions closed,
- mark theorem or lemma calls,
- mark the exact line where the conclusion is reached.

31 Second Pass: Set Algebra and Boolean Thinking

Set operations mirror logical operations.

$$x \in A \cap B \quad \Longleftrightarrow \quad x \in A \wedge x \in B.$$

$$x \in A \cup B \quad \Longleftrightarrow \quad x \in A \vee x \in B.$$

$$x \in A^c \quad \Longleftrightarrow \quad \neg(x \in A).$$

This is why set identities often resemble logic laws.

31.1 De Morgan's laws

$$(A \cup B)^c = A^c \cap B^c.$$

$$(A \cap B)^c = A^c \cup B^c.$$

These are set versions of:

$$\neg(P \vee Q) \Longleftrightarrow \neg P \wedge \neg Q.$$

$$\neg(P \wedge Q) \Longleftrightarrow \neg P \vee \neg Q.$$

31.2 Worked proof

Theorem 20.

$$(A \cup B)^c = A^c \cap B^c.$$

Proof. Let x be arbitrary. Then

$$x \in (A \cup B)^c$$

if and only if $x \notin A \cup B$. This holds if and only if it is not the case that $x \in A$ or $x \in B$. By De Morgan's law for logic, this means $x \notin A$ and $x \notin B$. Therefore $x \in A^c$ and $x \in B^c$, which means $x \in A^c \cap B^c$. Hence the sets are equal. □

31.3 Why this matters

Analysis uses set operations constantly: unions of open sets, intersections of closed sets, complements, limsup and liminf of sets. Probability uses sigma-algebras of sets. Topology is built from collections of open sets. Algebraic geometry defines closed sets as zero sets of polynomials and studies their complements as open sets.

31.4 Exercises

1. Prove $(A \cap B)^c = A^c \cup B^c$.
2. Prove $A \setminus B = A \cap B^c$.
3. Prove $A \subseteq B$ if and only if $B^c \subseteq A^c$.
4. Translate the previous item into an implication and its contrapositive.

32 Second Pass: Functions as Maps of Structure

Functions do more than assign values. In higher mathematics, functions preserve or reveal structure.

32.1 Maps in different fields

In analysis, continuous functions preserve nearness.

In algebra, homomorphisms preserve operations:

$$\phi(ab) = \phi(a)\phi(b).$$

In topology, homeomorphisms preserve topological structure.

In differential geometry, smooth maps preserve differentiable structure.

In algebraic geometry, morphisms preserve polynomial/algebraic structure.

The word "map" is not casual. Mathematics often studies objects by studying the structure-preserving maps between them.

32.2 Kernel as what a map forgets

In group theory, the kernel of a homomorphism $\phi : G \to H$ is

$$\ker \phi = \{g \in G : \phi(g) = e_H\}.$$

This set contains the elements that become invisible under ϕ.

In linear algebra, the null space of a matrix is the set of vectors sent to zero. Same idea.

32.3 Quotients as controlled forgetting

If a map forgets some information, a quotient often formalizes exactly what has been forgotten. The first isomorphism theorem says, roughly:

$$G/\ker \phi \cong \operatorname{im} \phi.$$

Translation: once you identify elements that the map cannot distinguish, what remains is essentially the image.

This is far beyond the first proof course, but the seed is already in functions, equivalence relations, and inverse images.

32.4 Exercise

For each function, say what it forgets.

1. $f : \mathbb{Z} \to \{0, 1\}$ where $f(n)$ is the parity of n.
2. $g : \mathbb{R}^2 \to \mathbb{R}$ where $g(x, y) = x$.
3. $h : \mathbb{R} \to \mathbb{R}$ where $h(x) = x^2$.
4. A hash function from strings to fixed-length hashes.

Walking question

Think of a function you use in software. What information does it preserve? What information does it discard? If two inputs have the same output, what relation does that define?

33 Second Pass: Relations, Classification, and Quotients

Equivalence relations classify objects. They say which objects belong in the same bucket.

33.1 Classification by invariant

An invariant is a property that stays the same under some relation or transformation.

For congruence modulo 5, the invariant is remainder after division by 5.

For similar triangles, the invariant is shape, not size.

For topological spaces up to homeomorphism, invariants include connectedness, compactness, and number of holes.

33.2 Equivalence classes

If $\sim$ is an equivalence relation on A, the equivalence class of a is

$$[a] = \{x \in A : x \sim a\}.$$

For congruence modulo 5:

$$[2] = \{\ldots, -8, -3, 2, 7, 12, 17, \ldots\}.$$

33.3 Partitions

Equivalence classes partition the set:

- every element belongs to some class,
- different classes are either identical or disjoint.

This is a theorem worth proving.

Theorem 21. *If $\sim$ is an equivalence relation on A, then any two equivalence classes are either disjoint or equal.*

Proof. Suppose $[a]$ and $[b]$ are not disjoint. Then there exists $x \in [a] \cap [b]$. Thus $x \sim a$ and $x \sim b$. By symmetry, $a \sim x$. Since $a \sim x$ and $x \sim b$, transitivity gives $a \sim b$.

We show $[a] = [b]$. Let $y \in [a]$. Then $y \sim a$. Since $a \sim b$, transitivity gives $y \sim b$, so $y \in [b]$. Thus $[a] \subseteq [b]$.

Similarly, using $b \sim a$, we get $[b] \subseteq [a]$. Therefore $[a] = [b]$. □

33.4 Why classification matters

Mathematics often classifies objects up to an equivalence relation:

- fractions up to equality of ratios,
- geometric figures up to congruence or similarity,
- groups up to isomorphism,
- spaces up to homeomorphism,
- manifolds up to diffeomorphism.

Classification is one of the great mathematical projects: understand all objects of a certain kind, considering two objects the same when they have the same relevant structure.

34 Second Pass: Induction Beyond Formulas

Induction is not only for sums. It is for any structure built step by step.

34.1 Recursive definitions

The Fibonacci numbers are defined by:

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+2} = F_{n+1} + F_n.$$

The definition itself is recursive. Proofs about Fibonacci numbers often use induction because the object is built inductively.

34.2 Worked Fibonacci proof

Theorem 22. *For all $n \geq 1$,*

$$F_1 + F_2 + \cdots + F_n = F_{n+2} - 1.$$

Proof. Base case: For $n = 1$, the left side is $F_1 = 1$, and the right side is $F_3 - 1 = 2 - 1 = 1$.

Induction step: Suppose

$$F_1 + \cdots + F_k = F_{k+2} - 1.$$

Then

$$F_1 + \cdots + F_k + F_{k+1} = (F_{k+2} - 1) + F_{k+1}.$$

Using the Fibonacci recurrence,

$$F_{k+2} + F_{k+1} = F_{k+3}.$$

Thus

$$F_1 + \cdots + F_{k+1} = F_{k+3} - 1.$$

This is the desired statement for $k + 1$. Therefore the formula holds for all $n \geq 1$. □

34.3 Structural induction preview

In computer science, data structures like lists and trees are recursive. To prove something about all lists, you prove it for the empty list and prove that adding an element preserves the property. To prove something about trees, you prove it for leaves and show it is preserved when subtrees are joined.

That is structural induction. It is the same spirit as ordinary induction but applied to recursively built objects.

34.4 Exercise

Write an induction proof for the length of concatenated lists:

$$\text{len}(L_1 ++ L_2) = \text{len}(L_1) + \text{len}(L_2).$$

State clearly what object you are inducting on.

35 Second Pass: Analysis as Quantifier Discipline

Analysis is where vague words like "approaches," "near," and "small" are put under oath.

35.1 The anatomy of an epsilon proof

To prove $a_n \to L$, the structure is:

1. Let $\varepsilon > 0$ be arbitrary.
2. Choose N depending on ε.
3. Let $n \geq N$.
4. Show $|a_n - L| < \varepsilon$.

The proof often has a scratch-work phase where you discover how to choose N.

35.2 Scratch work versus proof

For $a_n = 3/n$, we want:

$$\left|\frac{3}{n} - 0\right| < \varepsilon.$$

This is:

$$\frac{3}{n} < \varepsilon.$$

Solving for n gives:

$$n > \frac{3}{\varepsilon}.$$

That scratch work suggests choosing $N > 3/\varepsilon$.

Now the proof:

Proof. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ with $N > 3/\varepsilon$. If $n \geq N$, then

$$\left|\frac{3}{n}\right| = \frac{3}{n} \leq \frac{3}{N} < \varepsilon.$$

Thus $3/n \to 0$. □

35.3 Why scratch work is not the proof

Scratch work often runs backward from the desired conclusion. The final proof must run forward from the chosen N to the conclusion.

This distinction matters in all proof writing. Discovery and presentation are not identical.

35.4 Exercise

Prove:

$$\lim_{n\to\infty} \frac{5}{n+2} = 0.$$

First write scratch work. Then write the formal proof.

36 Second Pass: Algebra as Lawful Manipulation

Abstract algebra asks what follows from laws. The objects may be numbers, matrices, symmetries, functions, or something stranger. The laws are what matter.

36.1 Cancellation in groups

Theorem 23. *If G is a group and $ab = ac$, then $b = c$.*

Proof. Suppose $ab = ac$. Multiply on the left by a^{-1}:

$$a^{-1}(ab) = a^{-1}(ac).$$

By associativity,

$$(a^{-1}a)b = (a^{-1}a)c.$$

Since $a^{-1}a = e$,

$$eb = ec.$$

Since e is the identity,

$$b = c.$$

□

36.2 Why not divide?

In ordinary algebra, you might say "divide by a." In a group, the operation may not be multiplication of numbers, and it may not be commutative. The correct group-theoretic action is multiplying by the inverse on the correct side.

36.3 Noncommutativity

In many groups, $ab \neq ba$. Matrix multiplication is the standard example. Symmetries of a square also do not always commute: rotating then reflecting can differ from reflecting then rotating.

This is why order matters in algebraic proofs.

36.4 Exercise

Prove right cancellation: if $ba = ca$, then $b = c$.

Then explain why your proof must multiply by a^{-1} on the right rather than on the left.

37 Second Pass: Topology as the Study of Stable Nearness

Topology begins with a bold move: stop measuring distance directly and keep only the structure needed to talk about continuity.

37.1 Open sets as observable neighborhoods

In $\mathbb{R}$, an open interval like $(2, 5)$ has the property that each point inside has room to move. The point 2.1 has a small interval around it inside $(2, 5)$. The point 2 is not inside, so it does not matter.

In $[2, 5]$, the point 2 is inside but has no room to move left while staying inside. That is why $[2, 5]$ is not open in $\mathbb{R}$.

37.2 Relative openness

But $[2, 5]$ can be open as a subset of itself with the subspace topology. This is a major beginner surprise: openness depends on the ambient space.

In the space $X = [2, 5]$, the set $[2, 3)$ is open in X because it equals:

$$X \cap (1, 3).$$

The surrounding universe changed.

37.3 Why this matters

Manifolds are locally modeled on open subsets of $\mathbb{R}^n$. Algebraic geometry uses topologies where closed sets are solution sets of polynomial equations. Functional analysis studies topologies on spaces of functions. The word "open" is portable because it is defined structurally.

37.4 Exercise

In the subspace $X = [0, 1] \subseteq \mathbb{R}$, decide whether the following are open in X:

1. $[0, 1/2)$,
2. $(0, 1/2)$,
3. $(1/2, 1]$,

4. $\{0\}$.

For each open set, express it as $X \cap U$ for some open $U \subseteq \mathbb{R}$.

38 Second Pass: Geometry, Coordinates, and What Is Real

Coordinates are useful, but they are not the object itself.

38.1 The coordinate trap

A point on Earth can be described by latitude and longitude. But the point is not the coordinate pair. Different coordinate systems can describe the same point. Near the poles, familiar coordinates behave badly.

Differential geometry takes this seriously. A manifold is studied through charts, but the manifold is not any one chart.

38.2 Local versus global

The Earth is locally flat enough for a city map. Globally it is not flat. This local/global tension appears everywhere:

- locally Euclidean spaces can have global holes,
- local derivatives can assemble into global curvature,
- local coordinate formulas must agree on overlaps.

38.3 Transition maps

If two charts describe overlapping regions of a manifold, the transition map tells how to convert coordinates from one chart to the other. Smooth manifolds require these transition maps to be smooth.

This is where functions, inverse functions, open sets, and differentiability all meet.

38.4 Exercise

Explain in your own words:

1. why one flat map cannot represent Earth perfectly,
2. why local maps are still useful,
3. how this relates to the definition of a manifold,

4. why proof-based definitions are needed to prevent coordinate confusion.

39 Second Pass: Algebraic Geometry as Translation

Algebraic geometry is a translation project between equations and shapes.

39.1 From equation to shape

The polynomial

$$f(x, y) = x^2 + y^2 - 1$$

defines the circle:

$$V(f) = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 - 1 = 0\}.$$

39.2 From shape to functions

Instead of only asking which points lie on the circle, ask which polynomial functions on the plane become the same when restricted to the circle.

On the circle,

$$x^2 + y^2 = 1.$$

So the polynomial $x^2 + y^2$ behaves like the constant function 1 on that shape.

This suggests quotienting the polynomial ring by the relation generated by $x^2 + y^2 - 1$:

$$\mathbb{R}[x, y]/(x^2 + y^2 - 1).$$

That quotient is a first glimpse of the coordinate ring.

39.3 Why ideals are natural

If a polynomial vanishes on a shape, then any multiple of it also vanishes on the shape. If two polynomials vanish on the shape, their sum vanishes too. That is exactly the behavior of an ideal.

39.4 The same early tools return

- Sets: solution sets.
- Functions: polynomial maps.
- Equivalence relations: quotient rings.
- Ideals: algebraic encoding of equations.

- Topology: closed sets as varieties.
- Proofs: every translation must be justified.

39.5 Exercise

Let $V = V(y - x^2) \subseteq \mathbb{R}^2$, the parabola.

1. Explain why y and x^2 define the same function on V.
2. Explain why $y - x^2$ vanishes on V.
3. Give another polynomial that vanishes on V.
4. What information does the quotient $\mathbb{R}[x, y]/(y - x^2)$ seem to remember?

40 Guided Solutions and Commentary

This section is not for passive reading. Try the problems first. Then use these as models for how solutions breathe.

40.1 Solution: sum of two odd integers

Claim: The sum of two odd integers is even.

Proof. Let a and b be odd integers. Then there exist $m, n \in \mathbb{Z}$ such that $a = 2m + 1$ and $b = 2n + 1$. Therefore

$$a + b = (2m + 1) + (2n + 1) = 2m + 2n + 2 = 2(m + n + 1).$$

Since $m + n + 1 \in \mathbb{Z}$, the sum $a + b$ is even. □

Commentary: The proof opens the definition of odd for both numbers and closes the definition of even at the end.

40.2 Solution: composition of surjections

Claim: If $f : A \to B$ and $g : B \to C$ are surjective, then $g \circ f : A \to C$ is surjective.

Proof. Let $c \in C$ be arbitrary. Since g is surjective, there exists $b \in B$ such that $g(b) = c$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. Then

$$(g \circ f)(a) = g(f(a)) = g(b) = c.$$

Thus for arbitrary $c \in C$, there exists $a \in A$ such that $(g \circ f)(a) = c$. Therefore $g \circ f$ is surjective. □

Commentary: The proof works backward from the target c. Surjectivity of g gets you to some b, then surjectivity of f gets you to some a.

40.3 Solution: power set inclusion

Claim: If $A \subseteq B$, then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Proof. Suppose $A \subseteq B$. Let $S \in \mathcal{P}(A)$ be arbitrary. Then $S \subseteq A$. Since $A \subseteq B$, every element of S is an element of B, so $S \subseteq B$. Therefore $S \in \mathcal{P}(B)$. Hence $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. □

Commentary: The elements of $\mathcal{P}(A)$ are themselves sets. This is a type issue. To prove $S \in \mathcal{P}(B)$, prove $S \subseteq B$.

40.4 Solution: no smallest positive real

Proof. Suppose, for contradiction, that r is the smallest positive real number. Since $r > 0$, we have $r/2 > 0$. Also $r/2 < r$. This contradicts the assumption that r is the smallest positive real number. Therefore no smallest positive real number exists. □

Commentary: The contradiction is exact: $r/2$ is a positive real smaller than the alleged smallest positive real.

41 Final Exam: From Beginner to Ready

This is not a trick exam. It is a mirror. Use it to find what needs another pass.

41.1 Part 1: Logic

1. Define statement and predicate.
2. Give an example of a predicate whose truth depends on the domain.
3. Negate: $\forall x \in A, \exists y \in B,\ R(x, y)$.
4. Negate: if n is prime, then n is odd.
5. Explain why $\forall x \exists y$ is different from $\exists y \forall x$.

41.2 Part 2: Proofs

1. Prove that the sum of two odd integers is even.
2. Prove that if $a \mid b$ and $a \mid c$, then $a \mid (b - c)$.
3. Prove that if n^2 is odd, then n is odd.
4. Prove that there is no smallest positive real number.
5. Prove that $\sqrt{2}$ is irrational.

41.3 Part 3: Sets and functions

1. Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
2. Prove that if $A \subseteq B$, then $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.
3. Prove that composition of surjective functions is surjective.
4. Give a counterexample showing $f(S \cap T) = f(S) \cap f(T)$ can fail.
5. Prove $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.

41.4 Part 4: Relations and induction

1. Prove congruence modulo n is symmetric.
2. Prove subset inclusion is a partial order.

3. Prove by induction that $1 + 3 + \cdots + (2n - 1) = n^2$.
4. Prove by induction that $2^n \geq n + 1$ for $n \geq 1$.
5. Explain why strong induction is useful for prime factorization.

41.5 Scoring

If you can complete most of this with clear definitions and justified lines, start Velleman. If not, do not call it failure. You have found the exact floorboard that needs reinforcement.